Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/28/2021
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (80NSSC19K0231 ; NNX16AK21G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	07/01/2016	**End Date:**	06/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2022 per PI (Ed., 8/5/21) NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/12/20) NOTE: Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and MSFC personnel (Ed., 2/21/19)		
Key Personnel Changes/Previous PI:	2020 report: Elisa Torresani, Ph.D., will be CoInvestigator in summer 2020.		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University) Torresani, Elisa Ph.D. (San Diego State University)		
Grant/Contract No.:	80NSSC19K0231 ; NNX16AK21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 2/21/19): Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and Marshall Space Flight Center (MSFC) personnel. NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments," grant #NNX14AB31G with the same Principal Investigator. A flight research project is in progress investigating the gravity-dependent densification, distortion, microstructure development, and the related macroscopic shape evolution of multi-phase powder alloys during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as the basis for techniques applied to in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit particle-based fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing based on sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer finite element code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity are being performed using a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon. As of April 2021 all microgravity experiments are completed consisting of 49 samples. Of these, 7 samples have been returned to the research team and analyzed. An additional 28 samples have returned to Earth and are within the NASA system undergoing extraction from transport hardware. A final 14 samples are sintered but not returned from the International Space Station. Final delivery of remaining samples is expected in October 2021. Ground proof experiments are not scheduled as yet, but at least one run is requested by the research team.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is responsible for approximately $25 billion in annual component production. The products are applied to such diverse fields as metal cutting, oil well drilling, automotive engines, electric motors, computer heat sinks, jet engines, medical tools, skeletal implants, exercise monitors, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge in sintering since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. During that high temperature sintering the component strength drops considerably and is subject to gravity effects. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by sintering. We are adding the missing gravitational term to computer simulations, typically finite element analysis tools, for predicting final shape and size. New insight has been gained on the role of pore saturation in component shape loss, indicating that under-sintering in reduced gravity is the appropriate protocol for net-shape fabrication. A mathematical model is offered to predict system softening versus densification.
Task Progress:	A comprehensive report on findings has been completed in the form of a 139 page compilation of all findings. This summary report shows how gravity influences liquid phase sintering. Further, a general purpose publication "Tungsten Heavy Alloy Handbook" has been completed that integrates over 900 prior publications with the current findings to provide an integrated view on microgravity and gravity sintering. Central to the reports are findings from microgravity liquid phase sintered tungsten heavy alloy compacts. The microgravity compacts change ratios of W-Ni-Cu-Mn to adjust the solid-liquid ratio and the tungsten solubility in the liquid. However, most of the ground experiments are on W-Ni-Cu and W-Ni-Fe compositions. Liquid phase sintered in microgravity has been performed on the International Space Station using 7 cartridges, each with 7 different compositions. Parallel experiments from ground experiments use batch laboratory sintering (with various atmospheres), dilatometry, and thermal analysis. Together these studies help quantify the densification and distortion trajectory differences associated with microgravity sintering. Pending are ground tests that follow the temperature-time trajectories applied using the flight furnace. The microgravity experiments involve a sequence of hold times from the onset of densification, at about 1180°C, to more than an hour at 1210°C. All measurements on the samples are made after return to Earth. The results capture the sintering dynamics in terms of densification, microstructure evolution, component shape change (shrinkage), and distortion. Documents have been generated that compile background information relevant to the experiments, including details on alloy development, sample fabrication details, initial compact characteristics, flight equipment attributes, qualification testing, heating cycles and related finite element analysis for temperature distribution, targeting the comparison of microgravity and ground sintering results. It further documents the microgravity thermal cycles and calculation of an integral work of sintering metric as the basis for understanding the behavior. The integral work of sintering is applied as a means to combine heating, high temperature hold time and temperature, and cooling into a single parameter. An effective activation energy of 250 kJ/mol is applied to the solid-state portion of the heating cycle and 125 kJ/mol to the liquid phase portion of the heating cycle. Modeling efforts are positioned to embed this information in finite element codes to enable size and shape predictions in various gravitational fields. The W-Ni-Cu-Mn compositions are necessary due to an artificial temperature limit of 1200°C for the furnace on the International Space Station. Due to the low sintering temperature, addition of Mn and formation of MnO from the oxygen contained in the tungsten powder, novel microstructures are observed. Residual pores abound in the sintered samples. This porosity reshaped, enlarged, and held sufficient off-gassing vapor pressure to resist densification. Due to

	the capillary stress associated with the residual pores, the components remained stiff and resisted distortion. As pores fill with liquid, saturation increases, and capillarity is lost, leading to compact distortion on Earth. In the ground experiments, over-sintering is observed where longer times and higher temperatures induce a loss of density. This is common to systems that produce internal gas via reactions in the pores. Thus, vapor phase stability for the pores is a significant difficulty with respect to microgravity sintering. In this case, the difficulty is attributed to the manganese alloying addition forming oxides during sintering. The oxygen is released by tungsten during the solution-reprecipitation cycle that is responsible for densification and grain growth. We find that in the microgravity samples, incomplete densification contributes a capillary force sufficient to retard distortion. This is not related to the large gas filled pores (stable with just a few ppm of CO or other gas at the sintering temperature), but comes from the small intergranular pores. The next wave of microgravity research should focus on reaching full density. A better solution would be to enable higher temperatures while avoiding the manganese additions. Vacuum sintering at 1500°C for a W-Ni-Fe composition is recommended for future research.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Abstracts for Journals and Proceedings	Torresani E, German RM, Olevsky EA. "Gravity Influence on Sintering of BJ Components." Presented at the TMS 2021. 150th Annual Meeting and Exhibition, The Minerals, Metals and Materials Society, Virtual Meeting, March 15-18, 2021. TMS 2021 150th Annual Meeting & Exhibition Supplemental Proceedings. , Mar-2021
Articles in Peer-reviewed Journals	Torresani E, German RM, Huff H, Olevsky EA. "Influence of gravity on sintering of 3D-printed powder components." J Am Ceram Soc. 28 July 2021;105:131-146. https://doi.org/10.1111/jace.18056 , Jul-2021